Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

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CCBR

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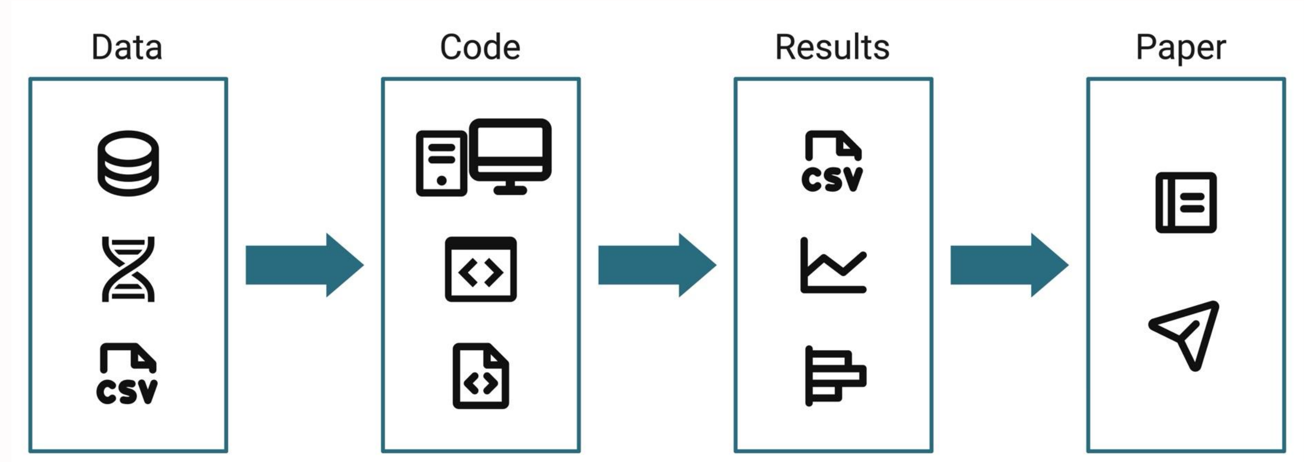
INTRODUCTION

Reproducibility is the foundation of trustworthy computational biology. Our framework operationalizes core principles across all pipelines



REPRODUCIBILITY
Same data; same methods

REPLICABILITY
Different data; same methods

ROBUSTNESS
Same data; different methods

GENERALIZABILITY
Different data; different methods

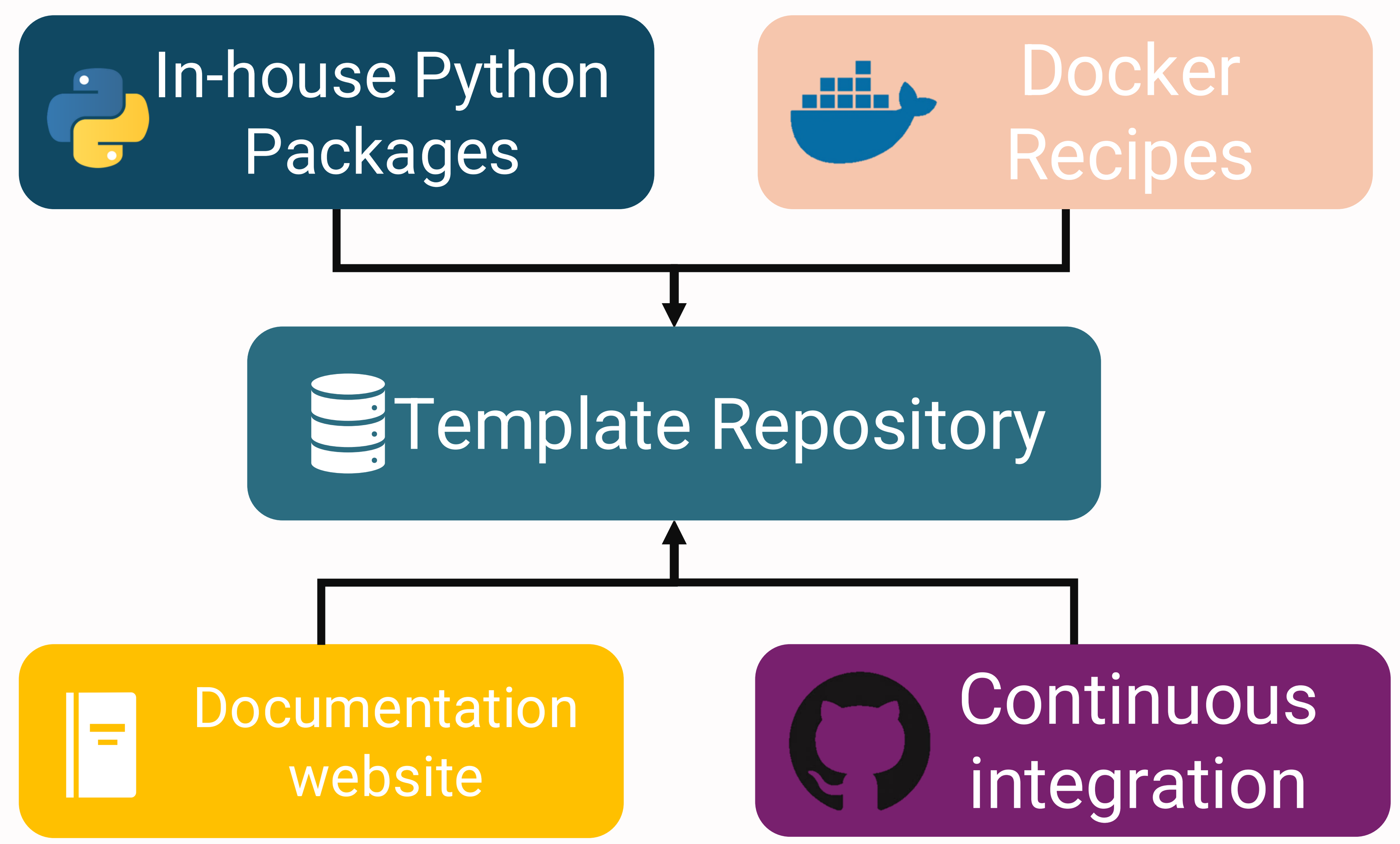
Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

- **Version Control**
- **Modular Design**
- **Automated Tests**
- **Documentation**

We enforce consistency through templates and automation. These practices reduces technical debt and improve sustainability

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



Unified framework from template to deployment with automated quality and delivery

- Fix once, deploy everywhere
- Consistent CLI
- Automated quality gates

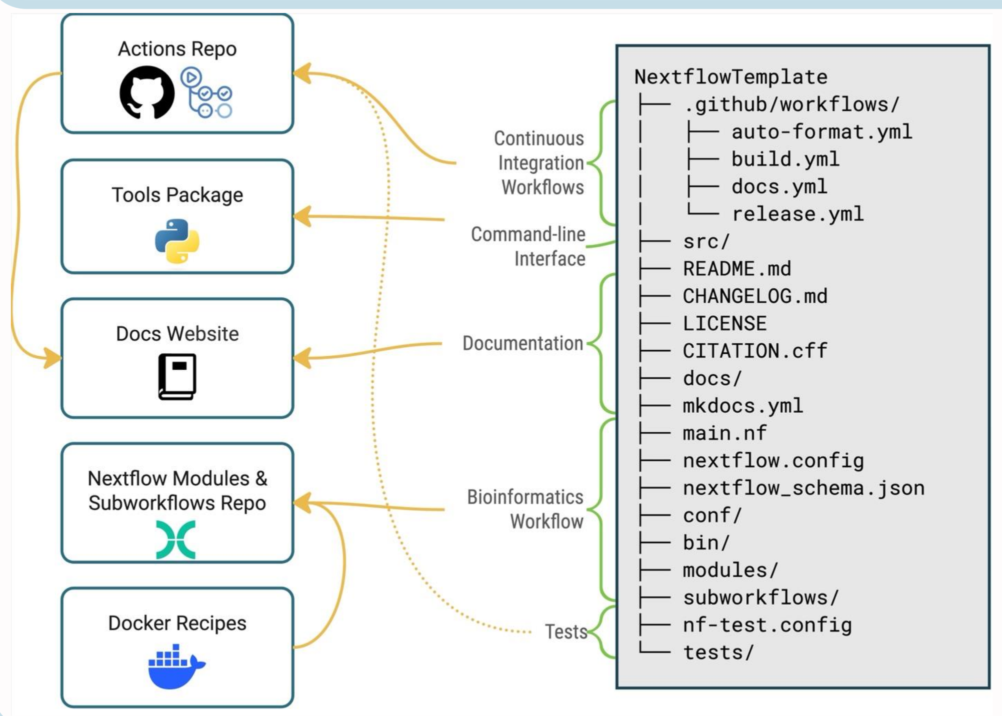
PIPELINES BUILT

Whole Genome **LOGAN**	ChIP-Seq **CHAMPAGNE**	scRNA-Seq **SINCLAIR**	CRISPR-Seq **CRISPIN**

10+ pipelines on NIH Biowulf HPC cluster

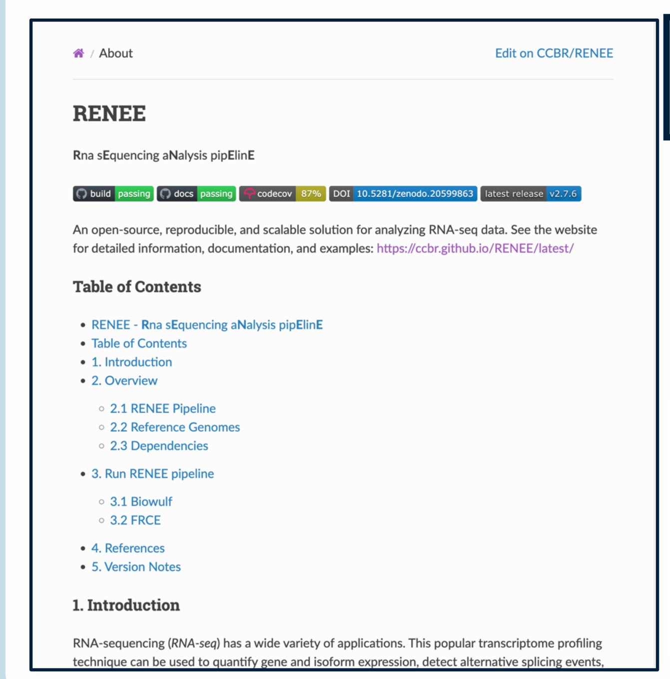
```
module load ccbrpipeliner
```

PIPELINE TEMPLATE



- Standardized directory structure for all pipelines
- Built in CI, testing, and containerization
- Easily extendible for newer methods and technologies

CLI, DOCUMENTATION, AUTOMATION



```
$ champagne run --input samplesheet
$ sinclair run --input samplesheet
```

Consistent documentation and orchestration across all pipelines simplifies usage and training

GitHub Actions runs CI checks on every change

Scalable, Reproducible, Sustainable Bioinformatics

NIH Biowulf HPC
Secure, Scalable, and high performance computing platform

GitHub Open Source
Collaborate, Contribute, Reuse

Documentation
Guides, Tutorials, Best Practices

CCBR GitHub

CCBR Website

Advancing reproducible computational research through standardized, containerized, and automated bioinformatics pipeline

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